Lab 7

Question 1 Practice (1-4)

Practice 1

```
student@Echo2: ~/wildcardpractice
student@Echo2:~$ curl -s https://cis106.com/assets/scripts/wildcardpractice.sh | bash
--2026-04-07 21:07:42--  https://cis106.com/assets/logo100x100.png
Resolving cis106.com (cis106.com)... 185.199.108.153
Connecting to cis106.com (cis106.com)|185.199.108.153|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3520 (3.4K) [image/png]
Saving to: ‘logo100x100.png’

logo100x100.png     100%[===================>]   3.44K  --.-KB/s    in 0s

2026-04-07 21:07:42 (25.2 MB/s) - ‘logo100x100.png’ saved [3520/3520]

student@Echo2:~$ cd wildcardpractice
student@Echo2:~/wildcardpractice$ ls random*
random_text.md  randomtext.txt
student@Echo2:~/wildcardpractice$ ls *.csv *.txt
78food.txt  _random.csv  randomtext.txt  Sample1234.csv  sample123.csv
student@Echo2:~/wildcardpractice$ ls random*.md
random_text.md
student@Echo2:~/wildcardpractice$ ls sample* *.md
random_text.md  sample1.doc  sample99.docx        sample.pdf
sample123.csv   sample1.xls  samplejsonfile.json  samplesite.html
student@Echo2:~/wildcardpractice$
```

Practice 2

```
student@Echo2: ~/wildcardpractice
student@Echo2:~$ cd wildcardpractice
student@Echo2:~/wildcardpractice$ ls *.???
2020-08-27-capture.raw  baik.xml   book.exe         randomtext.txt
2022-02-02-camera.jpg   beatk.exe  book.pdf         Sample1234.csv
78food.txt              beatk.pdf  book.xml         sample123.csv
allfiles.lst            beatk.xml  fail.log         sample1.doc
Allfiles.lst            bk.exe     file.log         sample1.xls
baik.exe                bk.pdf     logo100x100.png  sample.pdf
baik.pdf                bk.xml     _random.csv      Sample.pdf
student@Echo2:~/wildcardpractice$ ls .*
.bash_history  .hidden1.py   .hidden2.py
.hidden1.doc   .hidden2.doc  .hiddenFile.txt
student@Echo2:~/wildcardpractice$ ls b??*.???
baik.exe  baik.xml   beatk.pdf  book.exe  book.xml
baik.pdf  beatk.exe  beatk.xml  book.pdf
student@Echo2:~/wildcardpractice$ ls sample?.*
sample1.doc  sample1.xls
student@Echo2:~/wildcardpractice$ ls sample?.???
sample1.doc  sample1.xls
student@Echo2:~/wildcardpractice$ ls .*.??
.hidden1.py  .hidden2.py
student@Echo2:~/wildcardpractice$
```

Practice 3

```
1/1                                student@Echo2: ~/wildcardpractice
student@Echo2:~/wildcardpractice$ ls [a-z]*.???
allfiles.lst  beatk.pdf  book.exe  logo100x100.png  sample.pdf
baik.exe      beatk.xml  book.pdf  randomtext.txt
baik.pdf      bk.exe     book.xml  sample123.csv
baik.xml      bk.pdf     fail.log  sample1.doc
beatk.exe     bk.xml     file.log  sample1.xls
student@Echo2:~/wildcardpractice$ ls [A-Z][0-9].*
ls: cannot access '[A-Z][0-9].*': No such file or directory
student@Echo2:~/wildcardpractice$ ls [A-Z]*[0-9]*.*
Sample1234.csv
student@Echo2:~/wildcardpractice$ ls [0-9][0-9][0-9][0-9]*.*
2020-08-27-capture.raw      2022-02-02-camera.jpg
2021-04-10-screenshot.jpeg
student@Echo2:~/wildcardpractice$ ls [0-9][0-9][0-9][0-9]*.???
2020-08-27-capture.raw  2022-02-02-camera.jpg
student@Echo2:~/wildcardpractice$ ls [!A..E..I..O..U..a..e..i..o..u]
ls: cannot access '[!A..E..I..O..U..a..e..i..o..u]': No such file or directory
student@Echo2:~/wildcardpractice$ ls [!AEIOUaeiou]*
2020-08-27-capture.raw      bk.pdf           Sample1234.csv
2021-04-10-screenshot.jpeg  bk.xml           sample123.csv
2022-02-02-camera.jpg       book.exe         sample1.doc
78food.txt                  book.pdf         sample1.xls
baik.exe                    book.xml         sample99.docx
baik.pdf                    fail.log         samplejsonfile.json
baik.xml                    file.log         sample.pdf
beatk.exe                   logo100x100.png  Sample.pdf
beatk.pdf                   _random.csv      samplesite.html
beatk.xml                   random_text.md
bk.exe                      randomtext.txt
student@Echo2:~/wildcardpractice$ echo "Bruh..."
Bruh...
student@Echo2:~/wildcardpractice$ ls [!a-zA-Z0-9]*
_random.csv
student@Echo2:~/wildcardpractice$ echo "Professor, I don't know if you'll read this but I spend a good 15 minutes in another tab trying that one ^"
Professor, I don't know if you'll read this but I spend a good 15 minutes in another tab trying that one ^
student@Echo2:~/wildcardpractice$ =
```

Practice 4

```
1/1                                student@Echo2: ~
student@Echo2:~$ mkdir -p ~/music/{jazz,rock}/{mp3files,videos,oggfiles}/new
student@Echo2:~$ tree -a ~/music
/home/student/music
├── jazz
│   ├── mp3files
│   │   └── new
│   ├── oggfiles
│   │   └── new
│   └── videos
│       └── new
└── rock
    ├── mp3files
    │   └── new
    ├── oggfiles
    │   └── new
    └── videos
        └── new

15 directories, 0 files
student@Echo2:~$
```

Question 2 Challenge 1

```
1/1                                student@Echo2: ~/assets
student@Echo2:~$ mkdir -p ~/wallpapers/cars/{1080p,2k,4k}
student@Echo2:~$ cd wallpapers
student@Echo2:~/wallpapers$ tree -a
.
└── cars
    ├── 1080p
    ├── 2k
    └── 4k

5 directories, 0 files
student@Echo2:~/wallpapers$ cd
student@Echo2:~$ mkdir -p ~/assets/{imgs,video}/{large,small}
student@Echo2:~$ cd assets
student@Echo2:~/assets$ tree -a
.
├── imgs
│   ├── large
│   └── small
└── video
    ├── large
    └── small

7 directories, 0 files
student@Echo2:~/assets$
```

Question 3 Challenge 2

```
1/1                                student@Echo2: ~/website_ch1_lab7
student@Echo2:~$ curl https://cis106.com/assets/scripts/chq-lab7.sh | bash
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   395  100   395    0     0   3248      0 --:--:-- --:--:-- --:--:--  3264
student@Echo2:~$ cd website_ch1_lab7
student@Echo2:~/website_ch1_lab7$ tree -a
.
├── 404.html
├── api.js
├── APP.conf
├── APP.conf.bk
├── app.js
├── backup.sh
├── build.sh
├── cart.svg
├── DB1.conf.old
├── deploy.sh
├── footer.html
├── footer.jpg
├── header.html
├── hero.png
├── index.html
├── item1.raw
├── item2.jpg
├── navbar.html
├── post1.html
├── post2.html
├── post3.html
├── posts.json
├── _Products.Json
├── _reset.css
├── Robots.txt
├── sidebar.html
├── _style.css
├── _theme.css
└── Users.JSON

1 directory, 29 files
student@Echo2:~/website_ch1_lab7$ mkdir -p app assets/css assets/images components config/deploy data pages
student@Echo2:~/website_ch1_lab7$ tree -a
```

student@Echo2: ~/website_ch1_lab7

```
.
├── 404.html
├── api.js
├── app
├── APP.conf
├── APP.conf.bk
├── app.js
├── assets
│   ├── css
│   └── images
├── backup.sh
├── build.sh
├── cart.svg
├── components
├── config
│   └── deploy
├── data
├── DB1.conf.old
├── deploy.sh
├── footer.html
├── footer.jpg
├── header.html
├── hero.png
├── index.html
├── item1.raw
├── item2.jpg
├── navbar.html
├── pages
├── post1.html
├── post2.html
├── post3.html
├── posts.json
├── _Products.Json
├── _reset.css
├── Robots.txt
├── sidebar.html
├── _style.css
├── _theme.css
└── Users.JSON
```

student@Echo2: ~/website_ch1_lab7

```
student@Echo2:~/website_ch1_lab7$ mv _*.css assets/css
student@Echo2:~/website_ch1_lab7$ mv *.svg *.jpg *.png *.raw assets/images/
student@Echo2:~/website_ch1_lab7$ mv *.sh config/deploy/
student@Echo2:~/website_ch1_lab7$ mv *.json *.JSON data/
student@Echo2:~/website_ch1_lab7$ tree -a
.
├── 404.html
├── api.js
├── app
├── APP.conf
├── APP.conf.bk
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
│       ├── cart.svg
│       ├── footer.jpg
│       ├── hero.png
│       ├── item1.raw
│       └── item2.jpg
├── components
├── config
│   └── deploy
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
│   ├── posts.json
│   └── Users.JSON
├── DB1.conf.old
├── footer.html
├── header.html
├── index.html
├── navbar.html
├── pages
├── post1.html
├── post2.html
```

```
1/1                                   student@Echo2: ~/website_ch1_lab7
├── post3.html
├── _Products.Json
├── Robots.txt
└── sidebar.html

10 directories, 29 files
student@Echo2:~/website_ch1_lab7$ mv *.Json data/
student@Echo2:~/website_ch1_lab7$ tree -a
.
├── 404.html
├── api.js
├── app
├── APP.conf
├── APP.conf.bk
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
│       ├── cart.svg
│       ├── footer.jpg
│       ├── hero.png
│       ├── item1.raw
│       └── item2.jpg
├── components
├── config
│   └── deploy
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
│   ├── posts.json
│   ├── _Products.Json
│   └── Users.JSON
├── DB1.conf.old
├── footer.html
├── header.html
├── index.html
```

```
1/1                                   student@Echo2: ~/website_ch1_lab7
├── navbar.html
├── pages
├── post1.html
├── post2.html
├── post3.html
├── Robots.txt
└── sidebar.html

10 directories, 29 files
student@Echo2:~/website_ch1_lab7$ mv post*.html pages/
student@Echo2:~/website_ch1_lab7$ mv footer.html header.html navbar.html sidebar.html components/
student@Echo2:~/website_ch1_lab7$ mv APP.conf APP.conf.bk DB1.conf.old config/
student@Echo2:~/website_ch1_lab7$ tree -a
.
├── 404.html
├── api.js
├── app
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
│       ├── cart.svg
│       ├── footer.jpg
│       ├── hero.png
│       ├── item1.raw
│       └── item2.jpg
├── components
│   ├── footer.html
│   ├── header.html
│   ├── navbar.html
│   └── sidebar.html
├── config
│   ├── APP.conf
│   ├── APP.conf.bk
│   ├── DB1.conf.old
│   └── deploy
│       ├── backup.sh
```

```
1/1                                   student@Echo2: ~/website_ch1_lab7
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
│   ├── posts.json
│   ├── _Products.Json
│   └── Users.JSON
├── index.html
├── pages
│   ├── post1.html
│   ├── post2.html
│   └── post3.html
└── Robots.txt

10 directories, 29 files
student@Echo2:~/website_ch1_lab7$ mv api.js app
student@Echo2:~/website_ch1_lab7$ tree -a
.
├── 404.html
├── app
│   └── api.js
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
│       ├── cart.svg
│       ├── footer.jpg
│       ├── hero.png
│       ├── item1.raw
│       └── item2.jpg
├── components
│   ├── footer.html
│   ├── header.html
│   ├── navbar.html
│   └── sidebar.html
├── config
│   ├── APP.conf
```

```
1/1                                   student@Echo2: ~/website_ch1_lab7
├── config
│   ├── APP.conf
│   ├── APP.conf.bk
│   ├── DB1.conf.old
│   └── deploy
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
│   ├── posts.json
│   ├── _Products.Json
│   └── Users.JSON
├── index.html
├── pages
│   ├── post1.html
│   ├── post2.html
│   └── post3.html
└── Robots.txt

10 directories, 29 files
student@Echo2:~/website_ch1_lab7$ mv app.js app/
student@Echo2:~/website_ch1_lab7$ tree -a
.
├── 404.html
├── app
│   ├── api.js
│   └── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
│       ├── cart.svg
│       ├── footer.jpg
│       ├── hero.png
│       ├── item1.raw
│       └── item2.jpg
├── components
│   ├── footer.html
```

1/1 student@Echo2: ~/website_ch1_lab7

```
│   ├── api.js
│   └── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
│       ├── cart.svg
│       ├── footer.jpg
│       ├── hero.png
│       ├── item1.raw
│       └── item2.jpg
├── components
│   ├── footer.html
│   ├── header.html
│   ├── navbar.html
│   └── sidebar.html
├── config
│   ├── APP.conf
│   ├── APP.conf.bk
│   ├── DB1.conf.old
│   └── deploy
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
│   ├── posts.json
│   ├── _Products.Json
│   └── Users.JSON
├── index.html
├── pages
│   ├── post1.html
│   ├── post2.html
│   └── post3.html
└── Robots.txt

10 directories, 29 files
student@Echo2:~/website_ch1_lab7$
```